

1. What Reasoning Effort Changes

GPT-6 Astra supports configurable reasoning effort. In the API model reference the available levels are Low, Medium, High, XHigh, and Max. In Codex, the exact labels exposed by a surface can differ, but the practical control is the same: how much deliberative work the model is allowed to spend before and during execution.

A common mistake is to treat the scale as a quality slider with one universally best endpoint. That ignores cost, latency, task structure, and the fact that verification and tools can contribute more than extra internal reasoning to some problems.

2. A Simple Decision Model

Let $Q(e)$ be expected task quality at effort e , $C(e)$ the usage cost, and $T(e)$ the latency. The rational choice is not $\operatorname{argmax} Q$ by itself. It is an effort level whose expected marginal improvement is worth the added C and T for the consequences of this task.

For a reversible formatting fix with a deterministic test, a small gain in abstract reasoning has little value. For a difficult architectural migration with ambiguous constraints, an additional reasoning step may prevent hours of wrong work.

A practical policy:

1. Begin at Low or Medium when the task is well-specified, tool-rich, and cheaply verifiable.
2. Escalate when failures reflect planning or reasoning rather than missing context or broken tooling.
3. Prefer improving evidence, references, or tests when the agent simply lacks information.
4. Use the highest effort when the task's difficulty and consequence justify it, not because it exists.

3. Kundel's Observation

After extensive Astra use, Dominik Kundel reports that Low and Medium can be surprisingly strong, including in Blender work, and recommends trying them before assuming Max is necessary. This is an experience report, not a theorem. Its importance is methodological: compare outputs at the task level rather than inheriting an effort setting from an older model.

4. The Escalation Objection

Objection: Starting low can waste time. If a task is obviously hard, a failed Low run plus a High rerun may cost more than starting High.

Reply: Correct. The policy should condition on prior evidence. A fresh algorithmic proof, security-critical root-cause analysis, or broad refactor may already justify High or above. "Start low" is not a prohibition on judgment; it is a rejection of the assumption that maximum effort dominates every task.

5. Effort versus Environment

Suppose Astra cannot reproduce a browser bug because the environment lacks the user's authenticated state. Increasing reasoning does not create that state. Giving it Chrome may. Suppose it keeps violating an undocumented design convention. More reasoning may amplify the wrong inference. Supplying the actual design reference may solve the problem more directly.

This yields a diagnostic table:

Symptom | First lever

Missing facts | Context or tool access

Wrong durable rule | Audit AGENTS.md or skill

Shallow plan | Increase reasoning or request explicit planning

Flaky result | Improve reproduction and verification

Slow routine work | Lower effort or narrow scope

Cross-app blockage | Better tool routing

6. Changing Effort Midstream

Current model guidance supports changing reasoning effort in an ongoing conversation through configuration updates in supported API flows while preserving useful cached context. Conceptually, this suggests another policy: spend cheap effort on reconnaissance and deterministic edits; reserve high effort for the branch point where alternatives genuinely require difficult judgment.

The danger is fragmentation. A task should not oscillate effort after every uncertainty. Escalation needs a reason: a failed hypothesis, a consequential ambiguity, an adversarial review finding, or evidence that the current plan cannot account for observed behavior.

7. A Small Evaluation

To decide an effort default for your own work, choose ten representative tasks. Hold the prompt, repository state, and verification procedure constant. Compare Low, Medium, and High on completion rate, human corrections, elapsed time, and usage. Repeat enough tasks to avoid deciding from one spectacular demo. The point is not to discover a universal ranking; it is to estimate the frontier for your work.

References: OpenAI, GPT-6 Astra model reference; OpenAI, GPT-6 Astra model guidance; Dominik Kundel, "Five things I learned from using Astra."

8. Calibration by Failure Type

Suppose three Medium-effort runs fail. Run A misses a file that a simple repository search would reveal. Run B finds the files but chooses an architecture that violates a subtle invariant. Run C reaches the correct design but a tool crashes before the test completes. Only B is strong evidence for increasing reasoning effort. A is an evidence-gathering failure; C is an environment failure. Escalation should target the bottleneck actually observed.

9. Expected Value of Escalation

Let the extra cost of High over Medium be D . Let p be the probability that the harder reasoning changes a wrong decision into a correct one, and L the loss avoided if it does. Escalation is attractive when pL exceeds D plus the latency cost. No exact monetary estimate is required. The point is to expose the hidden variables: consequence and probability of benefit. A security-sensitive migration and a cosmetic copy edit should not share a default merely because they use the same model.

Independent Open Course

<https://github.com/dicnunz/codex-astra-course>

CODEX.00 | Codex with GPT-6 Astra

Handout 4: Is More Reasoning Better?

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